

L2: Case Study

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2026-09-21

Today's Agenda

- We will split you into groups of 3 to 4.
- You get an application for which you will write two user requirements.
- We discuss them in class.
- You get the requirements of another group and write 2 to 3 non-functional requirements for it.
- We discuss those in class.
- You get the user requirements of yet another group and write 1 functional requirement for it.
- We discuss those in class.

Metro Lahore Ticket Distributor Machine

- Metro lahore bus service wants a TicketDistributor machine that issues tickets for passengers travelling on buses and trams / light rail.
- Tickets for:
 - a single trip,
 - a round trip, or
 - multiple trips.
- (Discounted) passes for:
 - a day,
 - a week, or
 - a month.
- Different ticket rates for:
 - a child,
 - an adult, or
 - a senior person.

A Typical Transaction

- A user using the display interface to select the type and quantity of tickets.
- Then choosing a payment method of either cash, credit/debit card or a UL smartcard.
- The ticket or tickets are printed and dispensed to the user along with an optional receipt.
- The machine is required to distinguish valid notes from fake currency notes.
- The machine is also supposed to be able to read an existing pass issued by a metro counter or a machine, displaying its validity.

Additional Requirements

- System operated by a touch screen.
- In spite of some “heavy” components, we want the system to be fast and usable.
- Every transaction must be logged, including aborted transactions.
- The machine must be able to handle various exceptions, such as:
 - Aborting the transaction before complete
 - An insufficient amount of cash paid
 - A rejected credit/debit card
 - Insufficient money on a UL smartcard
 - Money return in case of a user-aborted transaction
 - Change return after successful cash transaction.

Additional Requirements

- Show:
 - Insufficient balance in a smartcard
 - Up-to-date information printed on tickets, e.g.,
 - Departure date and time
 - Purchase date and time
 - Price
 - Validity period
 - Where the ticket was issued
 - Destination station, if applicable.
- In case of an exception, an error message is to be displayed.

Finally

- The machine is required to work in a heavy load environment such as in the morning or evening time during weekdays.
- During weekends performance and efficiency should not get affected.
- We do not want user feedback after every development stage of the TicketDistributor machine but after every two stages, in order to save time.

Ready, Set, Go!

- Split into groups of 3 to 4.
- Write two user requirements.
- We discuss them in class.
- You get the requirements of another group and write 2 to 3 non-functional requirements for it.
- We discuss those in class.
- You get the user requirements of yet another group and write 1 functional requirement for it.
- We discuss those in class.

Functional Requirements (FR)

A. Ticket and Pass Selection

FR-01: The system shall allow a user to select a ticket type: single trip, round trip, or multiple trips.

FR-02: The system shall allow a user to select a pass type: day, week, or month.

FR-03: The system shall allow a user to select a passenger category: child, adult, or senior.

FR-04: The system shall apply the correct fare and discount based on ticket/pass type and passenger category.

FR-05: The system shall allow the user to select the quantity of tickets or passes.

FR-06: The system shall allow the user to select a destination station, where applicable.

FR-07: The system shall display the fare and total amount before payment.

FR-08: The system shall support multiple tickets or passes in a single transaction.

FR-09: The system shall display available payment methods before payment.

B. Payment Handling

FR-10: The system shall accept cash payment.

FR-11: The system shall accept credit/debit card payment.

FR-12: The system shall accept UL smartcard payment.

FR-13: The system shall validate currency notes and distinguish valid notes from fake notes.

FR-14: The system shall reject fake currency notes with an error message.

FR-15: The system shall accumulate inserted cash and display the amount paid and remaining amount.

FR-16: If cash is insufficient, the system shall allow the user to insert more money or cancel the transaction.

FR-17: If the user cancels a cash transaction, the system shall return the inserted money.

FR-18: After a successful cash transaction, the system shall return the correct change.

FR-19: The system shall read and authenticate credit/debit cards.

FR-20: The system shall handle rejected credit/debit cards with an error message and allow another payment method.

FR-21: The system shall read a UL smartcard and check its balance.

FR-22: The system shall deduct the fare from the UL smartcard after successful validation.

C. Pass Reading and Validation

FR-26: The system shall read an existing pass issued by a metro counter or another machine.

FR-27: The system shall display the validity of an existing pass.

FR-28: The system shall reject an invalid or expired pass with an error message.

D. Ticket Printing and Dispensing

FR-29: The system shall print the selected ticket(s) or pass(es).

FR-30: The system shall dispense the printed ticket(s) or pass(es) to the user.

FR-31: The system shall print an optional receipt if requested by the user.

FR-32: The system shall print up-to-date information on tickets, including departure date/time, purchase date/time, price, validity period, issuing location, and destination station if applicable.

FR-33: The system shall handle printer errors, paper jams, or empty paper and either retry, cancel, or refund the transaction.

FR-34: The system shall prevent duplicate ticket dispensing.

E. Touchscreen and User Interaction

FR-35: The system shall be operated through a touchscreen interface.

FR-36: The system shall provide clear on-screen instructions for each step.

FR-37: The system shall provide a cancel/abort option at any stage of the transaction.

FR-38: The system shall display an error message for every exception or failed operation.

FR-39: The system shall time out an inactive session and abort the transaction safely.

FR-40: The system shall allow the user to choose a language, such as English or Urdu.

F. Transaction Logging and Auditing

FR-41: The system shall log every transaction, including successful, failed, and aborted transactions.

FR-42: Each log entry shall include timestamp, transaction ID, ticket/pass details, amount, payment method, and status.

FR-43: The system shall log the reason for aborted or failed transactions.

FR-44: The system shall log cash replenishment, paper replacement, and maintenance events.

FR-45: The system shall generate audit reports for administrators.

G. Exception Handling

FR-46: If the user aborts before completion, the system shall return any inserted cash/card and not issue a ticket.

FR-47: If cash is insufficient, the system shall display an error and allow more cash or cancellation.

FR-48: If a credit/debit card is rejected, the system shall display an error and allow another payment method.

FR-49: If a UL smartcard has insufficient balance, the system shall display an error and allow another payment method.

FR-50: If the user aborts a cash transaction, the system shall return the money.

FR-51: After a successful cash transaction, the system shall return change.

FR-52: If a network or payment failure occurs, the system shall cancel or hold the transaction safely.

FR-53: If power fails, the system shall recover without losing committed transaction data.

H. Administration and Maintenance

FR-54: The system shall provide an admin login for maintenance staff.

FR-55: The system shall allow cash and paper replenishment.

FR-56: The system shall allow collection of cash and transaction logs.

FR-57: The system shall support remote diagnostics and software updates.

FR-58: The system shall allow configuration of fares, discounts, ticket types, and languages.

FR-59: The system shall monitor machine health and send alerts for faults.

Non-Functional Requirements (NFR)

A. Performance

NFR-01: 95% of ticket purchase transactions shall complete within 30 seconds during peak hours.

NFR-02: Touchscreen response time shall be less than 1 second.

NFR-03: Card authorization shall complete within 5 seconds under normal network conditions.

NFR-04: Ticket printing and dispensing shall complete within 5 seconds after payment.

NFR-05: The system shall handle heavy load during weekday morning and evening peak hours.

NFR-06: Weekend performance and efficiency shall not be affected.

NFR-07: The system shall support a defined minimum throughput, e.g., at least 300 passengers per hour.

NFR-08: System startup shall complete within 2 minutes.

NFR-09: Recovery after a minor failure shall complete within 5 minutes.

B. Usability

NFR-10: A first-time user shall be able to complete a purchase without staff assistance.

NFR-11: The interface shall use clear prompts, large buttons, and consistent navigation.

NFR-12: The system shall support English and Urdu.

NFR-13: The system shall be accessible to elderly and disabled users, e.g., high contrast, audio prompts, and tactile buttons.

NFR-14: Error messages shall be clear, non-technical, and actionable.

C. Reliability and Availability

NFR-15: The system shall have at least 99.5% uptime during operating hours.

NFR-16: MTBF shall be greater than 10,000 hours.

NFR-17: MTTR shall be less than 30 minutes.

NFR-18: The system shall tolerate faults in printer, cash acceptor, card reader, and network.

NFR-19: No committed transaction data shall be lost during power failure.

NFR-20: The system shall automatically recover to a safe state after failure.

D. Security

NFR-21: Card payments shall comply with PCI DSS.

NFR-22: Sensitive payment data shall be encrypted in transit and at rest.

NFR-23: Counterfeit note detection accuracy shall be at least 99%.

NFR-24: The cash box and transaction logs shall be tamper-resistant.

NFR-25: Admin access shall use role-based authentication.

NFR-26: The system shall prevent unauthorized software or hardware access.

E. Auditability and Logging

NFR-27: Every transaction, including aborted transactions, shall be logged.

NFR-28: Logs shall be tamper-evident and time-stamped.

NFR-29: Logs shall be retained according to legal and operational requirements.

NFR-30: Logs shall synchronize with a central server when network is available.

NFR-31: The system shall provide traceability from ticket sale to payment and dispensing.

F. Maintainability and Supportability

NFR-32: The system shall use modular hardware and software components.

NFR-33: Remote diagnostics and fault alerts shall be supported.

NFR-34: Cash and paper replenishment shall be possible within 10 minutes.

NFR-35: Software updates shall be installable without full system downtime.

NFR-36: Maintenance documentation and training materials shall be provided.

G. Scalability and Portability

NFR-37: New ticket types, fares, discounts, and languages shall be addable without major redesign.

NFR-38: Additional payment methods shall be supportable.

NFR-39: The system shall be deployable across multiple stations and machines.

NFR-40: The software shall be portable across approved hardware platforms.

H. Compatibility

NFR-41: The system shall support EMV, ISO, NFC, and UL smartcard standards.

NFR-42: The system shall recognize valid Pakistani currency denominations.

NFR-43: The system shall integrate with the metro central fare and logging system.

I. Environmental and Physical

NFR-44: The machine shall operate in Lahore weather conditions, including heat, dust, and humidity.

NFR-45: The system shall include UPS or surge protection for power fluctuations.

NFR-46: The machine shall be physically robust against vandalism and heavy public use.

J. Legal, Compliance, and Process

NFR-47: The system shall comply with State Bank of Pakistan payment regulations.

NFR-48: The system shall comply with data privacy and consumer protection laws.

NFR-49: The system shall meet relevant accessibility standards.

NFR-50: User feedback shall be collected after every two development stages, not after every stage, to save time.